

Chapter 3. **GROUPS**

3.2 Subgroups



Today we will talk about the most important subgroups of a group: its **CENTRE**

Definition 3.19. The **centre** $Z(G)$ of a group G is the set of elements of G that commutes with every element of G ; that is:

$$Z(G) = \{z \in G \mid zg = gz \text{ for all } g \in G\}.$$

Notice that G is abelian if and only if $Z(G) = G$.

PROPOSITION 3.20. The centre $Z(G)$ of a group G is a subgroup.

Proof. We will use the Subgroup Test.

$1 \in Z(G)$ because $1g = g = g1$ for all $g \in G$.

Suppose that $z, z' \in Z(G)$ and let us show that $zz' \in Z(G)$ and $z^{-1} \in Z(G)$.

$$(zz')g = z(z'g) \stackrel{z' \in Z(G)}{=} z(gz') = (zg)z' \stackrel{z \in Z(G)}{=} (gz)z' = g(zz') \text{ for all } g \in G.$$

$$zg = gz \Rightarrow (z^{-1}z)g = z^{-1}gz \Rightarrow g = z^{-1}gz$$

$$\Rightarrow gz^{-1} = z^{-1}g(zz^{-1}) = z^{-1}g \quad \blacksquare$$

Examples 3.21.

1. $Z(\mathbb{Z}_4) = \mathbb{Z}_4$ and $Z(K_4) = K_4$ because both groups are abelian.

2. $Z(S_3) = \{\varepsilon\}$

Recall that $S_3 = \{\varepsilon, \sigma, \sigma^2, \tau, \tau\sigma, \tau\sigma^2\}$, where $\sigma = (1\ 2\ 3)$ and $\tau = (1\ 2)$.

S_3	ε	σ	σ^2	τ	$\tau\sigma$	$\tau\sigma^2$
ε	ε	σ	σ^2	τ	$\tau\sigma$	$\tau\sigma^2$
σ	σ	σ^2	ε	$\tau\sigma^2$	τ	$\tau\sigma$
σ^2	σ^2	ε	σ	$\tau\sigma$	$\tau\sigma^2$	τ
τ	τ	$\tau\sigma$	$\tau\sigma^2$	ε	σ	σ^2
$\tau\sigma$	$\tau\sigma$	$\tau\sigma^2$	τ	σ^2	ε	σ
$\tau\sigma^2$	$\tau\sigma^2$	τ	$\tau\sigma$	σ	σ^2	ε

τ and σ^2 do not commute with z , then

$$\underline{z, \tau, \sigma^2 \notin Z(S_3)}. \quad \tau z = z\sigma^2 \neq z\tau$$

$$\begin{aligned} z\sigma^2 &= \tau z \\ \sigma^2 z &= z\tau \end{aligned}$$

$z\tau$ and $z\sigma^2$ do not commute:

$$\begin{aligned} (z\tau)(z\sigma^2) &= \tau \\ (z\sigma^2)(z\tau) &= \sigma^2 \end{aligned}$$

So, $z\sigma$ and $z\sigma^2 \notin Z(S_3)$.

From here we obtain that $Z(S_3) = \{e\}$